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{
  "questions": [
    {
      "question": "In trigonometric equations, the expression  $\sin^2(x) + \cos^2(x)$  equals _____.",
      "options": ["1", "0", "x", "pi"],
      "correct_answer": 1,
      "solution": "This is a fundamental trigonometric identity.  $\sin^2(x) + \cos^2(x)$  always equals 1, regardless of the value of x. Hence, Option (1) is the right answer."
    },
    {
      "question": "The shape of a molecule determined by the VSEPR theory depends on the _____ of the electron pairs around the central atom.",
      "options": ["repulsion", "attraction", "hybridization", "resonance"],
      "correct_answer": 2,
      "solution": "VSEPR theory states that the shape of a molecule is mainly determined by the repulsion between the electron pairs surrounding the central atom. Hence, Option (2) is the right answer."
    },
    {
      "question": "The moment of inertia is greater when mass is distributed _____ from the axis of rotation.",
      "options": ["closer", "further", "equally", "randomly"],
      "correct_answer": 2,
      "solution": "Moment of inertia increases as the mass moves further from the axis of rotation because it depends on the distance squared from the axis. Hence, Option (2) is the right answer."
    }
  ]
}

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